# CS 486/686
# Markov Decision Processes I

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Lecture 13

RN 17.1 · PM 9.5

# Learning goals

- Motivate modeling a decision problem as a **Markov decision process.**
- Describe the **components** of a fully-observable MDP.
- Explain why we use a **discounted** reward function.
- Define the **policy** of an MDP.
- Give examples of how the **reward function** shapes the optimal policy.

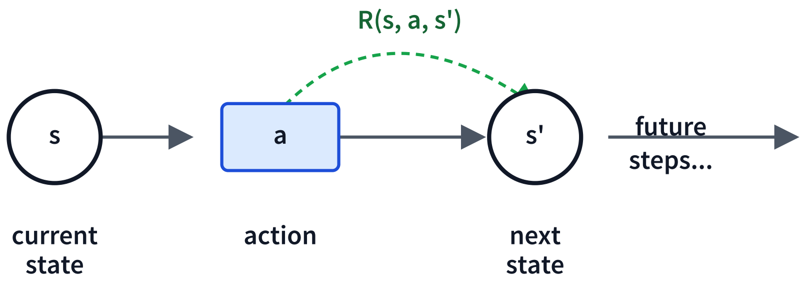
# From one-shot decisions to ongoing ones

## Two new things

- **Horizon:** the problem may never end.
  - *Infinite horizon:* the process may go on forever.
  - *Indefinite horizon:* the agent will stop eventually, but not at a known time.
- **Rewards along the way**, not just utility at the end.
  - The agent may never reach an "end".
  - Each step's reward includes action costs + bonuses/penalties.

A **Markov decision process (MDP)** formalizes this.

# A Markov decision process




**An MDP is a 4-tuple** $(S, A, P, R)$:

- $S$: set of states.
- $A$: set of actions.
- $P(s' \mid s, a)$: transition probabilities.
  *Stationary*: same for every $t$.
- $R(s, a, s')$: reward function.

For this lecture we'll use the simpler form $R(s)$ = reward for entering state $s$
.

# How do we aggregate rewards over time?

### Total reward

$\sum_{t=0}^{\infty} R(S_t)$

May diverge $\Rightarrow$ policies are not comparable.

### Average reward

$\lim_{n\to\infty} \frac{1}{n} \sum_{t=0}^{n} R(S_t)$

Zero whenever the total reward is finite.

### Discounted reward

$\sum_{t=0}^{\infty} \gamma^t R(S_t), \quad \gamma \in [0, 1)$

Always finite; the standard choice.

The discounted sum is also called the **return** $G(S_0)$.

# The discount factor $\gamma$

- **Time preference.** A dollar today is worth more than a dollar tomorrow.
- **Finiteness.** If $R(s) \in [-R_{\max}, R_{\max}]$ and $\gamma \in [0, 1)$:

$$\left|\sum_{t=0}^{\infty} \gamma^t R(S_t)\right| \le \frac{R_{\max}}{1-\gamma}.$$

- **Effective horizon.** Roughly $\frac{1}{1-\gamma}$ steps matter (e.g. $\gamma = 0.9 \Rightarrow$ about 10 steps).

## Variants

- **Fully-observable MDP** — the agent always knows its current state. (Today.)
- **Partially observable MDP (POMDP)** — combine MDP + HMM; the state is hidden behind observations.

# A 3 × 4 grid world

What should the robot do to maximize its rewards?

| | | | |
|---|---|---|---|
| Start | | | |
| | X | | −1 |
| | | | +1 |

- Let $s_{ij}$ be row $i$, column $j$. $s_{11}$ is the start.
- $s_{22}$ is a wall. $s_{24}$ and $s_{34}$ are goal states — the robot exits the world there.

# The grid world as an MDP

| | | | |
|---|---|---|---|
| Start | | | |
| | X | | −1 |
| | | | +1 |

**Actions:** up / down / left / right (in every state).

**Transition** $P(s' \mid s, a)$:

- 0.8 — action takes intended effect.
- 0.1 — 90° left of intended.
- 0.1 — 90° right of intended.
- If the robot bumps a wall, it stays put.

**Reward** $R(s)$: $R(s_{24}) = -1, R(s_{34}) = +1$; otherwise $R(s) = -0.04$.

# Understanding the transition model

**Q1.** The robot is in $s_{14}$ (top-right corner) and tries to move right. What is the probability it *stays* in $s_{14}$?

| | | | |
|---|---|---|---|
| Start | | | $s_{14}$ |
| | X | | −1 |
| | | | +1 |

A. 0.1

B. 0.2

C. 0.8

D. **0.9**

E. 1.0

**D — 0.9.** Trying right: 0.8 it bumps the wall (stays) + 0.1 it veers up (also bumps wall, stays) = 0.9. The remaining 0.1 veers down to $s_{24}$.

# Why a fixed action sequence is not enough

**Q2.** In a *deterministic* env, the sequence *down, down, right, right, right* is optimal for the grid world.

**True.** Five steps take us from $s_{11}$ to $s_{34}$.

**Q3.** In the *stochastic* env, the same sequence may end at *more than one* square.

**True.** E.g. five right-veers in a row send us to $s_{12}$.

**Q4.** So in the stochastic env, a fixed sequence is still good enough.

**False.** Actions misfire — we need to *re-plan* from whatever state we end up in.

| | | | |
|---|---|---|---|
| Start | | | |
| | X | | −1 |
| | | | +1 |

Need an action for **every state** — a **policy**.

# A policy

**Policy**

A **policy** $\pi$ specifies the agent's action as a function of the current state.

- **Stationary** — depends only on the state: $\pi : S \to A$.
- **Non-stationary** — also depends on the time step: $\pi : S \times \mathbb{N} \to A$.

For an infinite-horizon fully-observable MDP, the optimal policy is **stationary**.

# The reward function reshapes the optimal policy

Same dynamics; different $R(s)$ for non-goal states — very different optimal behaviors.

We will look at five qualitatively different reward regions:

1. $R(s) < -1.6284$ — *life is painful*
2. $-0.4278 < R(s) < -0.0850$ — *life is quite unpleasant*
3. $R(s) = -0.04$ — *life is unpleasant* (the default)
4. $-0.0221 < R(s) \leq 0$ — *life is only slightly dreary*
5. $R(s) > 0$ — *life is GOOD*

Each region balances **risk** (falling into −1) against **reward** (reaching +1 quickly).

# Five reward regions, five optimal policies

### painful

$R < -1.6284$

| | | | |
|---|---|---|---|
| → | → | → | ↓ |
| ↓ | X | → | −1 |
| → | → | → | +1 |

head to nearest exit, even −1

### quite unpleasant

$-0.43 < R < -0.085$

| | | | |
|---|---|---|---|
| ↓ | → | ↓ | ← |
| ↓ | X | ↓ | −1 |
| → | → | → | +1 |

shortcut through $s_{13}$

### unpleasant

$R = -0.04$

| | | | |
|---|---|---|---|
| ↓ | ← | ← | ← |
| ↓ | X | ↓ | −1 |
| → | → | → | +1 |

long way around; avoid −1

### slightly dreary

$-0.022 < R \leq 0$

| | | | |
|---|---|---|---|
| ↓ | ← | ← | ↑ |
| ↓ | X | ← | −1 |
| → | → | → | +1 |

no risk; bang against walls

### GOOD

$R > 0$

| | | | |
|---|---|---|---|
| ⊕ | ⊕ | ⊕ | ↑ |
| ⊕ | X | ← | −1 |
| ⊕ | ⊕ | ← | +1 |

wander forever; avoid both goals

Watch the action at $s_{13}$ (top, third column): it changes with every region.

# The value function

### Expected utility of a policy

- $V^\pi(s)$: expected utility of entering state $s$ and following $\pi$ thereafter.
- $V^\star(s)$: expected utility of entering $s$ and following the **optimal** policy $\pi^\star$.

| | | | |
|---|---|---|---|
| 0.705 | 0.655 | 0.611 | 0.388 |
| 0.762 | X | 0.660 | −1 |
| 0.812 | 0.868 | 0.918 | +1 |

$V^\star(s)$ for $\gamma = 1, R(s) = -0.04$ on non-goal states.

# The Q-function and the optimal policy

### Q-function

$$Q^\star(s,a) = R(s) + \textstyle\sum_{s'} P(s' \mid s,a)\, V^\star(s')$$

"If I were in $s$ and took action $a$ now, then followed $\pi^\star$ forever after — how good is my future?"

$$\pi^\star(s) = \arg\max_a Q^\star(s,a)$$

Given $V^\star$, the optimal policy is one greedy step away.

# Optimal action at $s_{13}$?

**Q5.** Using $Q^\star(s,a) = R(s) + \sum_{s'} P(s' \mid s,a) V^\star(s')$, what is $\pi^\star(s_{13})$?

| | | | |
|---|---|---|---|
| 0.705 | 0.655 | 0.611 | 0.388 |
| 0.762 | X | 0.660 | −1 |
| 0.812 | 0.868 | 0.918 | +1 |

$s_{13}$ is the yellow cell ($V^\star = 0.611$).

## Computing each Q-value (4 actions):

```
Left:  -0.04 + 0.8·0.655 + 0.1·0.611 + 0.1·0.660 = 0.6111
Right: -0.04 + 0.8·0.388 + 0.1·0.611 + 0.1·0.660 = 0.3975
Down:  -0.04 + 0.8·0.660 + 0.1·0.388 + 0.1·0.655 = 0.5923
Up:    -0.04 + 0.8·0.611 + 0.1·0.655 + 0.1·0.388 = 0.5531
```

**C — Left.** Left has the highest $Q^\star$ (0.6111). It moves toward the safer states; the −1 to the right is too close.

# Learning goals (recap)

✓ Motivate modeling a decision problem as a Markov decision process.

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✓ Explain why we use a **discounted** reward function.

✓ Define the **policy** of an MDP.

✓ Give examples of how the **reward function** shapes the optimal policy.

## Next: how do we compute $V^\star$?

Today: given $V^\star$, the optimal policy is just an argmax. But how do we *find* $V^\star$ in the first place?

L14: **Bellman equations** & **value iteration**.